Chapter 4
Interest Rates in the Risk-Free-Rate Era

Interest rates are the raw material of derivatives pricing. Every valuation in this book discounts something, and most of what we discount is itself a function of rates. It is worth spending a chapter on how rates are defined, measured, and assembled into a curve before we start using them.

A reader who learned interest rates from an older textbook will find this chapter organized differently from the way the subject was taught for the preceding thirty years. That is deliberate, and it reflects a genuine change in the market rather than a change in taste.

Until recently, the reference rate at the center of the derivatives market was LIBOR: a rate with a term structure built into it, quoted for one month, three months, six months, and a year, and fixed at the *start* of each interest period. Everything about how we taught interest rate derivatives assumed a rate of that shape. Forward rate agreements, floating-rate notes, caps, floors, and swaps were all defined against a rate you knew in advance.

LIBOR is gone. The rates that replaced it are **overnight** rates. They have no term structure of their own, they are published one day at a time, and a three-month interest amount computed from them is not known until the three months have elapsed. Every convention in the market had to be rebuilt around that fact, and a great deal of the machinery in Chapters 6, 7, 29, and 30 rests on getting these mechanics right.

This chapter therefore builds from the overnight rate upward. Sections 4.1 through 4.6 cover the rates themselves: what they measure, why the old ones were replaced, and how an overnight rate is compounded into the term rate a contract actually needs. Sections 4.7 through 4.12 cover the traditional apparatus (compounding conventions, zero curves, forward rates, duration, and convexity), which the transition left essentially intact. Readers who have met that apparatus elsewhere can move through the later sections quickly, but should not skip Sections 4.4 and 4.5, which have no counterpart in older treatments.

4.1 TYPES OF INTEREST RATES

An interest rate in any situation defines the amount of money a borrower promises to pay a lender. The higher the credit risk of the borrower, the higher the rate. Rates are quoted with a compounding frequency and a day count convention attached; ignoring either is a common and expensive error, and we return to both in Section 4.7. A rate also has a **tenor** (the length of the period it applies to, such as overnight, one month, or three months), which is not the same thing as the maturity of the instrument that pays it. A ten-year swap whose floating rate resets every three months has a maturity of ten years and a tenor of three months. The distinction runs through the whole book: the tenor is what the rate measures, the maturity is how long the exposure lasts.

Treasury rates. These are the rates at which a government borrows in its own currency. Because a government can in principle raise taxes or print money to repay such debt, these are conventionally treated as having negligible credit risk. Treasury rates are nevertheless *not* the rates derivatives markets discount at, for reasons that are partly technical and partly practical. Treasury yields differ from funding rates in either direction, depending on the supply of government debt, the balance-sheet cost of holding it, and the demand for it as collateral; Section 7.9 shows long dollar swap spreads negative since 2008, which puts Treasury yields above the swap curve. And a derivatives dealer cannot fund itself at the Treasury rate.

Repo rates. A repurchase agreement is a secured loan: one party sells securities and agrees to buy them back at a slightly higher price, the difference being the interest. The collateral need not be government securities. It can be almost any security the two sides agree on (agency mortgage-backed securities, corporate bonds, equities), and the rate depends on which it is. Better collateral commands a lower rate and a smaller **haircut** (the margin by which the collateral's market value exceeds the cash lent against it) because the lender of cash is less exposed if the borrower fails; repo against government bonds trades at the lowest rate, and repo against equities pays materially more. Credit risk is small but not zero, and it is a joint condition: the borrower must default *and* the collateral must have fallen in value before the lender loses anything.

Two further distinctions matter for what follows. A **general collateral** (GC) repo is one in which the lender of cash will take any security from an agreed class, and is therefore lending cash; a security in particular demand trades **special**, at a repo rate below general collateral and occasionally near zero, because the lender of cash is giving up interest for the privilege of borrowing *that* security. And repo trades at maturities from overnight out to several months. The dollar's reference rate is built from one narrow corner of all this (overnight, general collateral, US Treasury), and Section 4.2 explains why that corner and not another.

Overnight risk-free rates (RFRs). These are the successors to LIBOR and the subject of most of this chapter. Each is an average of actual overnight borrowing transactions in its currency, published once per business day by a central bank or exchange. The abbreviation **RFR** is used throughout this book, and in the market, for a rate of this kind.

Reference rates in derivatives. A **reference rate** is a published benchmark that contracts point at rather than negotiate individually. Its usefulness depends entirely on being both widely accepted and hard to manipulate. The story of the past decade is the market's discovery that its principal reference rate satisfied the first condition and not the second.

4.2 THE OVERNIGHT RISK-FREE RATES

Each major currency now has a designated overnight rate, selected by a working group of market participants and regulators: the Secured Overnight Financing Rate (SOFR) in dollars, the Sterling Overnight Index Average (SONIA), the Euro Short-Term Rate (€STR), the Tokyo Overnight Average Rate (TONA), and the Swiss Average Rate Overnight (SARON). Table 4.1 sets out how each is built.

Three features are common to all of them, and each is a direct response to a defect in LIBOR.

They are transaction-based. Each rate is computed from actual trades, not from estimates. SOFR, SONIA, and €STR are published the next morning from the previous day's transactions; SARON is computed in real time and fixed at the close of the same day from trades and binding

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Table 4.1: The principal overnight risk-free rates.

Currency	Rate	Basis	Administrator
USD	SOFR	Repo, US Treasuries	Federal Reserve Bank of New York
GBP	SONIA	Unsecured	Bank of England
EUR	€STR	Unsecured	European Central Bank
JPY	TONA	Unsecured	Bank of Japan
CHF	SARON	Repo, SNB GC basket	SIX Swiss Exchange

quotes; TONA has a same-day preliminary figure. SOFR is a volume-weighted median of overnight Treasury repurchase transactions and rests on well over $2 trillion of daily activity. LIBOR, at the end of its life, was in some tenors supported by a small number of transactions per day or none at all.

They are overnight. There is no one-month or three-month SOFR fixing in the way there was a one-month or three-month LIBOR. This is the feature with the largest practical consequences, and Section 4.4 is devoted to it.

They are close to risk-free. An overnight loan to a creditworthy counterparty carries almost no credit risk, and a secured overnight loan carries almost none even if the counterparty is weak. LIBOR, by contrast, was the rate at which *banks* could borrow unsecured for a term, and therefore embedded bank credit risk. That embedded credit component behaved unhelpfully: it widened exactly when the financial system was under stress, which is exactly when the contracts referencing it mattered most.

The policy rate is not the benchmark

A reader who knows the US money market will object that the table omits an obvious overnight dollar rate: the **effective federal funds rate** (EFFR), at which banks lend reserves to one another, published by the same New York Fed that publishes SOFR. It is not the benchmark, for two reasons worth separating.

Volume. Fed funds trading collapsed after 2008. Once reserves became abundant banks no longer needed to borrow them, and most of what remains is Federal Home Loan Banks (which cannot earn interest on reserve balances) lending to institutions that can. Daily volume runs on the order of $100 billion against SOFR's more than $2 trillion. A benchmark carrying hundreds of trillions of notional should not rest on the narrower market, which is the defect that ended LIBOR in a different form.

A policy rate makes a poor contract rate. EFFR is the rate the Federal Open Market Committee sets a target range for and steers toward, and it moves when policy decides it should. A benchmark ought to describe the market rather than be administered by an authority pursuing objectives of its own.

That second property is what keeps fed funds useful elsewhere. Being the policy rate makes it the rate in which expectations of policy are expressed (fed funds futures are how the market prices the next meeting), and the **fed funds–SOFR basis**, the spread between the two rates, trades actively through swaps that exchange one rate for the other, because a bank funds itself in one market and runs its derivative book in the other. The two rates can diverge with neither market under credit stress, for instance when Treasury collateral is scarce and repo rates fall relative to unsecured ones. Section 7.4 describes the October 2020 switch of dollar discounting from EFFR to SOFR, and the compensation it required.

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Secured versus unsecured

Table 4.1 shows that the rates are not all built the same way, and the distinction matters. SOFR and SARON are *secured*: they measure the cost of borrowing against collateral. SONIA, €STR, and TONA are *unsecured*: they measure the cost of borrowing overnight without collateral, and so retain a very small credit component.

The two secured rates are not secured against the same thing. SOFR is Treasury repo and nothing else. SARON is computed from the Swiss repo market against the general collateral basket of the Swiss National Bank (SNB), which admits cantonal and foreign government bonds, bonds of supranational institutions (for example, the European Investment Bank and the World Bank), and covered bonds alongside Confederation debt. "Secured" describes the structure of the loan, not the identity of the collateral, and a benchmark's designers choose both.

Which repo rate? A repo market is not one rate, and turning it into a benchmark takes choices. SOFR is drawn from three segments of the overnight Treasury market (tri-party repo, the interdealer General Collateral Finance market, and cleared bilateral transactions) and is a volume-weighted *median* rather than a mean, so that a small number of extreme trades cannot move it. The cheapest quarter of the bilateral volume is discarded before the median is taken, and that is how specials are kept out: a trade struck at a low rate because the lender wanted one particular bond says something about that bond, not about the price of cash. The same data supports two further published rates, the Tri-party General Collateral Rate and the Broad General Collateral Rate, which are repo rates but are not benchmarks. Not every repo rate is a reference rate, and what separates them is these choices rather than anything in the underlying trades.

It follows that SOFR is a proxy for the risk-free rate of an arbitrage argument rather than the thing itself. What a given firm can actually finance a given security at depends on that security and on that firm, which is where Section 5.10 picks the question up.

Being secured is not the same as being immune to stress. A secured rate reflects conditions in the collateral market, and those conditions can move violently for reasons having nothing to do with credit.

Case Study 4.1 September 2019: when the risk-free rate spiked

On September 17, 2019, SOFR fixed at 5.25%, up from 2.43% the day before. Some overnight repo trades executed at rates approaching 10% intraday. Nothing had happened to the creditworthiness of any borrower; the collateral in question was US Treasury securities.

What had happened was a collision of cash flows. A large corporate tax payment date coincided with the settlement of a heavy Treasury auction, draining reserves from the banking system at the moment dealers needed to finance a large inventory of new securities. Demand for overnight cash overwhelmed supply, and the price of cash, the repo rate, went to whatever level cleared the market. The Federal Reserve intervened with repo operations within a day.

The episode carries two lessons for anyone using SOFR. First, a rate can be free of *credit* risk and still be exposed to *funding* and *liquidity* conditions; "risk-free" is a statement about default, not about volatility. Second, the effect was concentrated in a single day. A contract referencing an overnight rate compounded over three months would have absorbed the spike into an average and barely noticed it. That is a real, if incidental, argument for the compounded-in-arrears conventions of Section 4.4. Predictable calendar effects of the same kind, milder but recurring, appear at each quarter-end and year-end.

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Negative rates

A feature of these rates that would have seemed impossible when LIBOR was designed is that they can be, and have been, negative. €STR was negative from its launch until 2022, following the European Central Bank's deposit facility rate below zero from 2014. TONA was negative from 2016 until the Bank of Japan ended its negative rate policy in March 2024, and SARON was negative from 2015 until 2022 and has fixed slightly below zero again since the SNB cut its policy rate to zero in June 2025.

This is not a curiosity. A lognormal model of rates cannot represent a negative rate, so when European rates went below zero the market for rate options moved to **normal** (or Bachelier) and **shifted lognormal** volatilities, a change Chapter 29 takes up.

A second consequence is contractual. Many floating-rate loans carry a zero floor, so the borrower never receives interest. A floor is an option, and a loan with an embedded floor is a loan plus a short position in that option, a position many borrowers did not realize they had sold until it moved into the money. It showed in hedges. A borrower that had swapped a zero-floored loan to fixed with an unfloored swap found that, once the floating rate went negative, the loan's floating payment stopped at zero while the swap's floating leg kept falling, so the borrower paid the fixed rate plus the size of the negative rate. The hedge leaked, which is why such hedgers bought floors to close the gap.

4.3 WHY LIBOR ENDED

LIBOR was, for nearly four decades, the most important number in finance. At its peak it was referenced by contracts with a notional value in the hundreds of trillions of dollars: swaps, futures, floating-rate notes, syndicated loans, student loans, and adjustable-rate mortgages.

Its construction was simple and, as it turned out, fatally so. Each business day a panel of banks was asked a question: at what rate could you borrow funds, were you to do so by asking for and then accepting interbank offers in a reasonable market size? The highest and lowest quarter of the submissions were discarded and the rest averaged, and the result published as that day's fixing for each currency and tenor.

The question is hypothetical. It asks what a bank *could* borrow at, not what it *did* borrow at. During normal conditions the distinction was immaterial. Two developments made it fatal.

The submissions were manipulable, and were manipulated. Because a bank's own derivatives book was sensitive to the fixing, a submitter had a direct financial interest in the answer. Beginning in 2012, investigations across several jurisdictions established that traders at multiple panel banks had requested and obtained submissions favorable to their positions, and that some banks had understated their submissions during the financial crisis to avoid signaling funding difficulty. The resulting fines exceeded $9 billion and several individuals were prosecuted, though several of the convictions were later overturned on appeal (by the US Court of Appeals for the Second Circuit in 2022 and the UK Supreme Court in 2025).

The underlying market disappeared. More fundamentally, banks largely stopped funding themselves through term unsecured interbank borrowing after the crisis. By the late 2010s, some LIBOR tenors were supported by only a small number of transactions in a day. A benchmark that hundreds of trillions of dollars of contracts depend upon cannot rest on the expert judgment of panels of 11 to 16 banks per currency answering a hypothetical question about a market that no longer trades.

In July 2017 the UK Financial Conduct Authority (FCA) announced that it would not compel panel banks to make submissions beyond the end of 2021, which amounted to setting a date for LIBOR's death. Most settings ceased at the end of 2021. The remaining and most

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important ones (overnight, one-, three-, six-, and twelve-month US dollar LIBOR) ceased on June 30, 2023. A temporary "synthetic" version (no longer a bank submission but term SOFR plus the fixed spread adjustment described below) was published for the one-, three-, and six-month US dollar settings only, for use in legacy contracts only, until September 30, 2024; the sterling and yen synthetic settings had ended earlier.

Not every interbank offered rate was retired. The Euro Interbank Offered Rate (EURIBOR), reformed in 2019 to a hybrid methodology that anchors submissions in transactions where they exist, remains the principal floating index for euro loans and swaps, and the Tokyo Interbank Offered Rate (TIBOR), Australia's bank bill swap rate (BBSW), the Hong Kong Interbank Offered Rate (HIBOR), and the Shanghai Interbank Offered Rate (Shibor) all continue. The Canadian Dollar Offered Rate (CDOR) ceased at the end of June 2024 in favor of the Canadian Overnight Repo Rate Average (CORRA). Where such a rate survives, it sits beside an RFR rather than in place of one, and the two-curve arithmetic of Section 4.10 applies to it directly.

Fallbacks: how existing contracts moved

Announcing that a rate will stop is straightforward. Moving the existing stock of contracts is not. A thirty-year swap written in 2015 against three-month LIBOR needs an answer to the question of what it references in 2030.

The answer for derivatives was the **ISDA 2020 IBOR Fallbacks Supplement and Protocol** (IBOR being the generic name for LIBOR-type interbank offered rates), which took effect in January 2021 and which the great majority of the market adhered to. It specifies that on cessation, a contract referencing LIBOR instead references

$$\text{compounded RFR in arrears} \;+\; \text{a fixed spread adjustment}$$

The spread adjustment is necessary because LIBOR contained bank credit risk and the RFR does not; simply substituting one for the other would transfer value from one side of every contract to the other. The adjustment was set as the five-year historical median of the spread between LIBOR and the compounded RFR, and, crucially, was fixed permanently on March 5, 2021, the date the FCA announced the cessation timetable. Table 4.2 gives the US dollar values.

Table 4.2: ISDA fallback spread adjustments for US dollar LIBOR, fixed March 5, 2021.

LIBOR tenor	Spread adjustment (bp)
1 month	11.448
3 months	26.161
6 months	42.826
12 months	71.513

Fixing the spread on a single date was the decisive design choice. It made every legacy contract's future economics knowable in advance, which is what allowed the transition to happen without a wave of disputes. It also means those numbers are now permanent features of the legacy book: a three-month dollar LIBOR swap that fell back in 2023 will reference SOFR plus 26.161 basis points for the rest of its life, regardless of what bank credit spreads do. Example 4.1 follows one such swap through a single payment.

Example 4.1 A legacy swap after fallback

A \$50 million interest rate swap written in 2015 pays three-month USD LIBOR quarterly. Following cessation, its floating leg references compounded SOFR in arrears plus the fixed adjustment of 26.161 basis points.

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For an accrual period of 92 days over which compounded SOFR works out to 4.28%, the floating rate is

$$4.28\% + 0.26161\% = 4.54161\%$$

and, on an actual/360 basis, the interest amount is

$$\$50{,}000{,}000 \times 0.0454161 \times \frac{92}{360} = \$580{,}317$$

Had LIBOR survived and fixed at 4.55% for the same period, the payment would have been \$581,389, a difference of about \$1,072, roughly 0.2% of the coupon.

That difference is deliberate. The spread adjustment was set to match the historical median gap between the two rates over five years; it was never intended to reproduce LIBOR period by period. What fixing the adjustment secures is that its level was known in advance and beyond either party's influence: whatever value it transferred between the two sides was fixed in advance and therefore priceable, which was the condition needed for the transition to proceed without renegotiating millions of contracts.

The protocol covered derivatives, whose counterparties could adhere to it. Cash products (loans, floating-rate notes, securitizations) carry their own fallback language, and many older US contracts had none that worked once LIBOR stopped. The Adjustable Interest Rate (LIBOR) Act of March 2022 and the Federal Reserve's implementing Regulation ZZ moved those contracts by statute, replacing US dollar LIBOR with a SOFR-based rate plus the same ISDA spread adjustment; for most cash contracts the replacement rate is CME term SOFR, the forward-looking rate of Section 4.6.

4.4 COMPOUNDING AN OVERNIGHT RATE IN ARREARS

We now come to the mechanical heart of the chapter. A contract needs an interest amount for a period (a quarter, say). The reference rate supplies one number per business day. How do we get from one to the other?

The answer adopted almost universally is to **compound the daily rates over the interest period**. The economic logic is that of rolling an overnight deposit: lend for one day at today's rate, reinvest principal and interest tomorrow at tomorrow's rate, and continue to the end of the period. The compounded rate is whatever single rate would have produced the same terminal value.

Formally, for an interest period running from T_1 to T_2 and containing d_b business days and d_c calendar days, with day-count base B:

$$R \;=\; \left[\prod_{i=1}^{d_b}\left(1 + \frac{r_i\, n_i}{B}\right) - 1\right] \times \frac{B}{d_c} \tag{4.1}$$

where r_i is the overnight rate published for business day i and n_i is the number of calendar days that rate applies for. On most days $n_i = 1$; on a Friday $n_i = 3$, because Friday's rate accrues across the weekend; around holidays it can be larger. The base depends on the currency: SOFR and €STR use actual/360, so $B = 360$, while SONIA and TONA use actual/365, so $B = 365$.

Two things about (4.1) deserve emphasis. First, the product is taken over *business* days, each rate weighted by the calendar days it covers, but the final scaling is by *calendar* days; this is what makes weekends accrue correctly. Second, and more consequentially:

> *R is not known until the last business day of the interest period.* A payment based on it cannot be calculated at the start of the period, only at the end. This is what "in arrears" means, and it is the single largest operational difference between the RFR world and the LIBOR world.

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Figure 4.1 contrasts the two arrangements. Under LIBOR a single fixing at T_1 determined the entire coupon, so both parties knew the amount for the whole period. Under an RFR the coupon accumulates one day at a time and is complete only at T_2—which is why every convention in Section 4.5 exists.

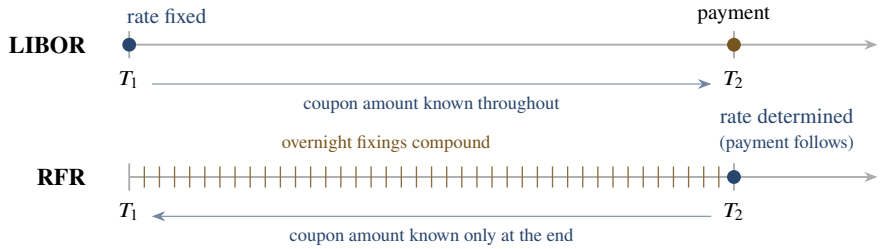


Figure 4.1: A rate set in advance against a rate determined in arrears.

Example 4.2 Compounding SOFR over one week

Consider a one-week interest period running from Monday the 6th to Monday the 13th: seven calendar days containing five business days, with Friday's rate applying for three days. SOFR accrues on actual/360, so $B = 360$.

Business day i	Date	r_i	n_i
1	Monday 6	4.31%	1
2	Tuesday 7	4.30%	1
3	Wednesday 8	4.32%	1
4	Thursday 9	4.33%	1
5	Friday 10	4.30%	3

The daily accrual factors are $1 + r_i n_i/360$:

$$1.000119722, \quad 1.000119444, \quad 1.000120000, \quad 1.000120278, \quad 1.000358333$$

Their product is 1.000838036, so from (4.1)

$$R = (1.000838036 - 1) \times \frac{360}{7} = 0.000838036 \times 51.428571 = 4.3099\%$$

For comparison, the simple day-weighted average of the same fixings is

$$\frac{4.31 + 4.30 + 4.32 + 4.33 + 3 \times 4.30}{7} = \frac{30.16}{7} = 4.3086\%$$

The compounded rate exceeds the simple average by 0.13 basis points over one week. The gap widens with the period: at a 4% rate level, compounding daily over a full year adds about 8 basis points relative to simple accrual. Over a quarter it is about 2 basis points: small, but not small enough to ignore in a market that quotes to a tenth of one. The gap grows roughly with the square of the rate level and in proportion to the term, so it is negligible when rates are near zero and becomes material quickly as they rise.

Published averages and indices

Because the arithmetic in Example 4.2 is tedious and error-prone, administrators publish it. The Federal Reserve Bank of New York publishes 30-, 90-, and 180-day compounded SOFR averages, and a **SOFR Index** that accumulates the daily compounding from a fixed start date. The index makes the compounded rate between any two dates a two-lookup calculation:

$$R = \left[\frac{\text{Index}(T_2)}{\text{Index}(T_1)} - 1\right] \times \frac{B}{d_c} \tag{4.2}$$

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where, as in (4.1), T_1 and T_2 are the start and end of the interest period, d_c is the number of calendar days between them, and $B = 360$ for SOFR. This is worth knowing not as a shortcut but because it eliminates a class of dispute: two parties reading the same two index values cannot disagree about the interest amount, whereas two parties each compounding sixty daily fixings can and do.

4.5 CONVENTIONS THAT MAKE COMPOUNDING WORKABLE

The problem with a rate known only on the final day of a period is that payment systems need notice. A borrower with a coupon due on the last day of a quarter cannot wait until the close of business that day to learn the amount; by then it is too late to arrange the payment.

The market's response was a family of conventions, each trading a small amount of economic precision for a usable notice period. A practitioner needs to recognize all of them, because a contract's value depends on which one it uses.

Payment delay. The simplest fix: compound over the full interest period, then pay a few business days after it ends. It is economically the cleanest, since the observation window matches the accrual window exactly. Its cost is a small timing adjustment, treated in Chapter 30.

Lookback. The rate for each day of the accrual period is taken from a fixed number of business days earlier, typically five. The final observation therefore occurs five days before period end, giving the notice required. In a **lookback without observation shift**, each observed rate is weighted by the calendar days of the accrual date it is applied to; in a **lookback with observation shift**, it is weighted by the calendar days of the observation date. The distinction sounds pedantic and is not. Without the shift, each rate is weighted by days that belong to a different date, a weighting mismatch: the compounded rate can no longer be reproduced from discount factors or from the SOFR Index, so it is not exactly the rate the curve prices. With the shift the observation period is a genuine compounding period, which is why the shifted convention is generally preferred.

Lockout (rate cutoff). The rate is observed normally until a cutoff a few days before period end, after which the last observed rate is frozen and repeated for the remaining days. It is simple to operate, at the cost of ignoring genuine rate movements in the final days, an awkward property if the cutoff spans a central bank meeting.

Last reset. This abandons in-arrears determination entirely, using the *previous* period's compounded rate as the current period's coupon. The borrower knows the coupon at the start of the period, as under LIBOR. The cost is that the rate is paid one period after it is determined, a timing adjustment priced with the machinery of Section 30.2 (Chapter 30 discusses this case at the end of Section 30.4).

Table 4.3 sets them side by side, writing k for the lookback, lockout, or delay length in business days, typically two to five. The column that matters operationally is the third: how much notice the payer gets. The column that matters for valuation is the fourth.

Figure 4.2 shows the same information as windows on a timeline, which makes the family resemblance clearer than the table can: each convention is the accrual period with its observation window slid, truncated, or displaced. Payment delay observes exactly the accrual period and defers only the payment. A lookback slides the whole window earlier. A lockout truncates it and repeats the last observed rate across the pale segment. Last reset abandons the current period altogether and takes the coupon from the previous one.

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Table 4.3: Conventions for compounding an overnight rate in arrears.

Convention	Observation window	Notice before payment	Principal cost
Payment delay	Matches the accrual period exactly	k days	Timing adjustment (Section 30.2)
Lookback, no observation shift	Shifted back k days; weights from accrual dates	k days	Weighting mismatch; rate not reproducible from discount factors or the index
Lookback with observation shift	Shifted back k days; weights from observation dates	k days	Minimal; observation period length may differ slightly from accrual
Lockout	Accrual period, but the final rate is frozen for the last k days	k days	Ignores rate moves late in the period, awkward across a policy meeting
Last reset	The *previous* accrual period	Full period	Coupon reflects a stale rate; timing adjustment (Section 30.2)

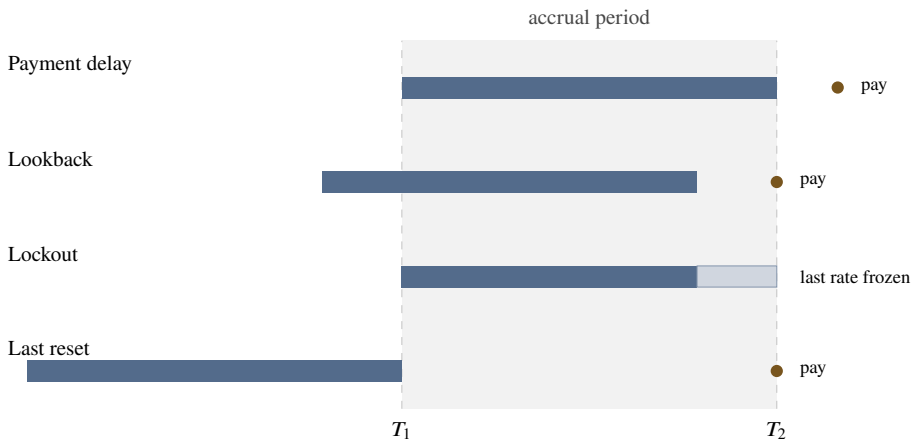


Figure 4.2: Observation windows (solid bars) against the accrual period (shaded).

The choice among these is not merely operational. Each convention implies a slightly different exposure, and a portfolio hedged on the assumption of one convention while transacted under another carries a small unhedged basis.

Cleared SOFR swaps use payment delay, usually two business days, and the interdealer market is uniform in this. Floating-rate notes use a lookback: five banking days with observation shift is the sterling standard for SONIA notes, while dollar notes have used both the shifted and unshifted forms, and the first agency SOFR notes of 2018 and 2019 paid a simple average of daily SOFR with a lockout before the market settled on compounding. Loans and securitizations, constrained by what a servicing system can compute, use lookbacks or fall back on last reset, or escape the question altogether by referencing the term rate of Section 4.6. The convention the Alternative Reference Rates Committee (ARRC), the market group that steered the dollar transition, recommended for SOFR loans is **daily simple SOFR**, which accrues each day's rate without compounding and so lets a borrower repay or draw down on any day without recomputing the period's interest.

Futures draw a distinction of their own. Three-month SOFR futures settle against daily SOFR *compounded* over the reference quarter, while one-month contracts settle against a

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simple arithmetic average over the calendar month. The underlying rate is the same and the accumulation is not, which is part of what the basis between the two contracts prices.

The division between derivatives and loans is not accidental. The derivatives market could rebuild its infrastructure once and centrally, through its clearing houses, and so adopted the convention that is cleanest economically. The loan market took whatever its existing servicing systems could support, fragmented across lookbacks and last reset, and in the end reached for a forward-looking term rate instead—which is where the next section begins.

4.6 TERM RATES AND CREDIT-SENSITIVE RATES

Two categories of rate sit alongside the compounded overnight rates, and both exist because some users genuinely wanted what LIBOR provided.

Term SOFR

A **term rate** is a forward-looking rate for a period, known at the start of that period (LIBOR's defining feature). Term SOFR is constructed not from submissions but from the SOFR derivatives market: CME publishes one-, three-, six-, and twelve-month term rates derived from SOFR futures prices, which embed the market's expectation of the average overnight rate over each period.

This is a sounder construction than LIBOR's, because it rests on traded prices. But it carries a structural vulnerability, which is circularity: term SOFR is built from derivatives that reference overnight SOFR, so it is only as robust as the volume of those derivatives. If exposures migrated from overnight SOFR to the term rate, with hedges following them, the market the term rate is computed from would thin while the exposures depending on it grew, which is the failure mode that ended LIBOR.

For that reason the ARRC recommended term SOFR for a deliberately restricted set of uses: business loans, and derivatives entered into to hedge those loans. It explicitly did *not* recommend it for the general derivatives market. The LIBOR Act added a third use by statute, as the replacement rate for legacy cash contracts (Section 4.3). The result is a two-tier structure, with loans referencing a forward-looking term rate and the derivatives market referencing compounded overnight rates, a basis that must itself be managed.

Credit-sensitive rates

A bank that funds itself unsecured and lends at a rate tied to SOFR has a genuine problem: in a crisis its funding costs rise while SOFR, being risk-free, does not. LIBOR's embedded credit component hedged that exposure automatically.

Several credit-sensitive benchmarks were launched to fill the gap, of which the most prominent was Bloomberg's Short-Term Bank Yield Index (BSBY). Regulators, including the SEC chair, questioned publicly whether it rested on sufficient transaction volume, and raised the concern that it reproduced LIBOR's central defect: a thin underlying market supporting enormous referencing volume, with the added feature that the thinness would be worst precisely during the stress the index was supposed to measure. Bloomberg announced its discontinuation in 2023 and ceased publication in November 2024. Smaller credit-sensitive indices such as Ameribor continue on a modest scale.

The episode is worth remembering when evaluating any proposed benchmark. The question is never only whether the index measures the right economic quantity. It is whether the market underneath it is deep enough to survive being referenced at scale, and whether it stays deep when conditions deteriorate.

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4.7 MEASURING INTEREST RATES

We now turn to the traditional apparatus, beginning with a point that causes more confusion than any other: a quoted rate is meaningless without its compounding frequency.

Suppose \$100 is invested for one year at 10%. With annual compounding the terminal value is \$110. With semiannual compounding it is $100 \times 1.05^2 = \$110.25$; with quarterly compounding $100 \times 1.025^4 = \$110.38$; with monthly, \$110.47. The same quoted number describes four different investments.

In general, an amount A invested for n years at rate R with compounding m times per year grows to

$$A\left(1 + \frac{R}{m}\right)^{mn} \tag{4.3}$$

Letting $m \to \infty$ gives **continuous compounding**, under which the terminal value is

$$A\, e^{Rn} \tag{4.4}$$

Continuous compounding is used throughout the option-pricing chapters of this book, because it makes the algebra dramatically simpler: discount factors multiply as $e^{-R_1 t_1} e^{-R_2 t_2} = e^{-(R_1 t_1 + R_2 t_2)}$, with no adjustment terms. It is a mathematical convenience, not a market convention—no instrument actually pays interest continuously.

Converting between conventions is essential and straightforward. Equating (4.3) and (4.4), a rate R_m with compounding frequency m corresponds to a continuously compounded rate

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right) \tag{4.5}$$

and conversely

$$R_m = m\left(e^{R_c/m} - 1\right) \tag{4.6}$$

Example 4.3 applies each in turn.

Example 4.3 Converting compounding frequencies

Consider a rate of 4% per annum with semiannual compounding. From (4.5), the equivalent continuously compounded rate is

$$R_c = 2\ln(1.02) = 2 \times 0.019803 = 3.9605\%$$

Conversely, a rate of 4% with continuous compounding corresponds, from (4.6), to

$$R_2 = 2\left(e^{0.02} - 1\right) = 2 \times 0.020201 = 4.0403\%$$

with semiannual compounding. The continuously compounded rate is always the lower of the two, since more frequent compounding achieves a given terminal value with a smaller quoted rate.

Day count conventions. A rate quotation is also incomplete without a day count, which specifies how a partial year is measured. The interest for a period is the notional times the rate times a year fraction, and the day count sets that fraction:

$$\text{interest} = N \times R \times \frac{\text{days in the period}}{\text{days in the year}},$$

with both counts defined by the convention. Four are in common use:

- **Actual/360**: actual days over 360. US dollar and euro money-market instruments and floating legs, the overnight rates SOFR and €STR, the annual fixed legs of swaps on SOFR and €STR, and China's Shibor;

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- **Actual/365 (fixed)**: actual days over 365. Sterling (SONIA), the overnight rates TONA and CORRA, Japanese government bonds, and Chinese repo, swaps on the seven-day repo fixing FR007, and renminbi fixed legs;
- **Actual/actual**: actual days over the actual length of the year or coupon period. US Treasury coupons and most other government bonds;
- **30/360**: every month counts as 30 days and the year as 360. US corporate bonds, and the fixed legs of euro swaps and of the older US dollar LIBOR swaps.

The difference is not trivial: an actual/360 rate of 4% earns interest on 365 days of a normal year, giving an actual/365-equivalent of $4\% \times 365/360 = 4.0556\%$. Mismatched day counts across the legs of a trade are a persistent source of small, systematic pricing errors.

4.8 ZERO RATES, BOND PRICING, AND BOOTSTRAPPING

The n-year **zero rate** (or spot rate) is the rate earned on an investment made today that pays nothing until year n. It is the term structure's fundamental object: every other rate is built from it.

A coupon-bearing bond's price is the sum of its cash flows discounted at the zero rate appropriate to each:

$$B = \sum_{i=1}^{n} c_i \, e^{-R_i t_i} \tag{4.7}$$

where c_i is the cash flow at time t_i and R_i the continuously compounded zero rate for maturity t_i. The bond's **yield** is the single rate y that, used for all maturities in (4.7), reproduces the observed price. The **par yield** is the coupon rate that makes the bond price equal to par.

Bootstrapping the OIS curve

Zero rates are not directly observable beyond the shortest maturities; they must be extracted from traded instruments. In the current market the natural instruments are **overnight indexed swaps** (OIS), swaps exchanging a fixed rate for the compounded overnight rate, because these are the most liquid and are what collateralized derivatives discount at.

The extraction procedure, **bootstrapping**, works outward from the shortest maturity, using each newly determined discount factor to solve for the next.

Example 4.4 Bootstrapping a zero curve from OIS rates

Suppose annual-paying OIS rates are quoted as 4.00%, 4.20%, and 4.30% for one, two, and three years. Because a swap has zero value at inception, and its floating leg plus a final notional payment is worth par (Section 7.3 shows why), its fixed rate (the swap's **par rate**) is by construction a par yield: each quote says that a bond paying that coupon prices at par.

One year. A single payment of 1.04 at $t = 1$ must be worth 1, so

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad z_1 = 4.0000\%$$

Two years. The two-year par bond pays 0.042 at $t = 1$ and 1.042 at $t = 2$:

$$0.042 \times 0.961538 + 1.042\,\mathrm{DF}_2 = 1 \implies \mathrm{DF}_2 = \frac{1 - 0.040385}{1.042} = 0.920936$$

so $(1 + z_2)^2 = 1/0.920936 = 1.085852$ and $z_2 = 4.2042\%$.

Three years.

$$0.043\,(0.961538 + 0.920936) + 1.043\,\mathrm{DF}_3 = 1 \implies \mathrm{DF}_3 = \frac{1 - 0.080946}{1.043} = 0.881164$$

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so $(1+z_3)^3 = 1.134863$ and $z_3 = 4.3072\%$.

At one year the zero and par rates coincide, since there is only one payment; beyond it each zero rate sits slightly above the corresponding par rate, which is the general pattern for an upward-sloping curve.

In practice the procedure is complicated by instruments that do not mature on convenient dates, by the need to interpolate between them, and by the requirement that the resulting curve be smooth enough that forward rates implied from it are not absurd. Curve construction is a specialty in its own right, and the choice of interpolation method is a genuine source of valuation differences between dealers.

4.9 FORWARD RATES AND FORWARD RATE AGREEMENTS

Forward rates are the future rates implied by today's zero curve. If investing for T_2 years must give the same result as investing for T_1 years and then reinvesting the proceeds at the forward rate until T_2, then the forward rate for the period $[T_1, T_2]$ satisfies

$$(1+z_1)^{T_1}(1+F_a)^{T_2-T_1} = (1+z_2)^{T_2}, \qquad 1+F\tau = \frac{\mathrm{DF}_1}{\mathrm{DF}_2} \tag{4.8}$$

where z_1 and z_2 are the annually compounded zero rates to T_1 and T_2, DF_1 and DF_2 the corresponding discount factors, and $\tau = T_2 - T_1$. The two forms state the same forward with different compounding: F_a is compounded annually, and F is a simple rate, the convention forward rate agreements use and the one meant below by "the forward rate." They coincide only for a one-year period, as in the applications below; otherwise $1+F_a = (1+F\tau)^{1/\tau}$. In continuously compounded terms the relationship (4.9) is cleaner still:

$$e^{R_1 T_1} \times e^{F(T_2-T_1)} = e^{R_2 T_2}, \qquad \text{so} \qquad F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} \tag{4.9}$$

where R_1 and R_2 are the continuously compounded zero rates and F here denotes the continuously compounded forward.

Applying (4.8) to Example 4.4 gives the one-year forward rate starting in one year, and the one-year rate starting in two years:

$$\frac{0.961538}{0.920936} - 1 = 4.4088\%, \qquad \frac{0.920936}{0.881164} - 1 = 4.5136\%.$$

Note how much more steeply the forward curve rises than the zero curve, as Figure 4.3 shows: zero rates are averages of forwards, so a modestly rising zero curve implies a considerably steeper forward curve. This amplification is a recurring feature and a frequent source of surprise.

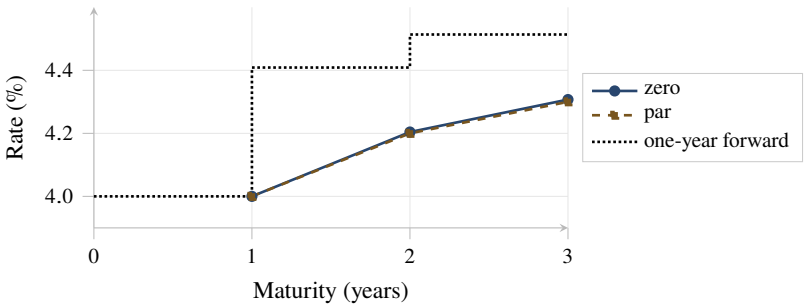


Figure 4.3: Par, zero and one-year forward rates from Example 4.4.

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Forward rate agreements. A forward rate agreement (FRA) is an agreement to exchange a fixed rate for a floating rate on a notional principal for a future period. Its value follows from a single observation: an FRA is worth zero when the fixed rate equals the forward rate. More generally, an FRA in which the fixed rate R_K is received in exchange for the realized rate, with notional L over accrual τ, is worth

$$V = L\,\tau\,(R_K - F)\,\mathrm{DF}_2 \tag{4.10}$$

to the party receiving fixed, where F is the current forward rate. Equation (4.10) is the simplest instance of the general principle developed in Chapter 30: we value the contract by substituting the forward rate for the unknown future rate, an operation whose validity is not automatic and whose exceptions occupy that chapter.

An FRA traditionally settles at the start of the period, T_1, as soon as the rate has fixed, rather than at T_2 when the interest would be due. The payment is therefore the period's interest difference discounted back over the period at the realized rate R: the party paying fixed receives $L\,\tau\,(R - R_K)/(1 + R\tau)$ at T_1, a negative amount meaning it pays.

FRAs were a major product in the LIBOR era, when a single fixing determined the settlement, and they still trade where a term rate survives: banks and corporate borrowers use EURIBOR FRAs to hedge the next few resets on euro loans. In the RFR currencies their role has narrowed considerably, since a short OIS achieves the same exposure with conventions that match the rest of the market. The short OIS market is where dealers, banks, and funds position for central-bank decisions, often through meeting-dated OIS whose accrual periods run from one policy meeting to the next.

4.10 THE MULTI-CURVE REALITY

One assumption made implicitly above needs flagging, because Chapter 7 dismantles it.

We have spoken of "the" zero curve, as though a single term structure describes a currency. Before 2008 that was approximately true. It is not true now, for a simple reason: discounting a collateralized derivative correctly requires the rate paid on the collateral, which under a standard credit support annex is the overnight rate. Meanwhile the rate a contract's cash flows are *indexed* to may be something else: EURIBOR (the largest surviving interbank rate), term SOFR, a foreign currency's RFR, or a legacy fallback rate.

The consequence is that valuation requires at least two curves: a discount curve built from OIS, and one or more forecast curves for the indices the contract references. Their difference is not noise. Cross-currency basis spreads (the additional spread required to swap one currency's RFR against another's) have at times exceeded 50 basis points and are themselves traded.

For the present chapter the single-curve treatment is adequate and keeps the mechanics visible. Chapter 7 develops the multi-curve framework properly, and treats it as the standard case rather than as a correction to a single-curve baseline.

4.11 DURATION AND CONVEXITY

We close with the two standard measures of a bond's interest rate sensitivity, both of which recur throughout the book. Convexity in particular is the subject of Chapter 30.

Duration measures the sensitivity of a bond's price to a parallel shift in the zero curve. With continuous compounding, if the bond has price B and cash flows c_i at times t_i, its duration is the cash-flow-weighted average time to payment:

$$D = \frac{\sum_{i=1}^{n} t_i\, c_i e^{-y t_i}}{B} \tag{4.11}$$

from which

$$\Delta B \approx -B D\,\Delta y \tag{4.12}$$

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When rates are quoted with compounding frequency m rather than continuously, the corresponding sensitivity uses **modified duration**, $D^* = D/(1 + y/m)$.

Equation (4.12) is a first-order approximation and inherits the usual limitation: it is a straight line fitted to a curve. The curvature is **convexity**,

$$C = \frac{1}{B}\frac{\partial^2 B}{\partial y^2} \tag{4.13}$$

giving the improved approximation

$$\Delta B \approx -BD\,\Delta y + \frac{1}{2}B\,C\,(\Delta y)^2 \tag{4.14}$$

Since $C > 0$ for an ordinary bond, the convexity term is positive for a rate move in either direction: a bondholder gains more when rates fall than they lose when rates rise by the same amount. Convexity is therefore desirable, and, as Chapter 30 shows at length, it is priced.

Figure 4.4 shows why. The duration approximation is the tangent to the price–yield curve at the current yield, and because that curve is convex the tangent lies *below* it on both sides. Duration alone therefore understates the price whichever way rates move; the shaded region between the two lines is what the convexity term in (4.14) recovers.

Two features of the figure repay attention. The bond is the one of Example 4.5. Plotting the percentage change rather than the price level makes the tangent's slope exactly minus the modified duration (here −2.7429 per percentage point of yield), independent of the bond's price, so the picture illustrates the definition rather than one particular bond. And the yield range shown is deliberately far wider than anything a three-year bond would realistically see: over a plausible range the two lines are almost indistinguishable, which is precisely why duration alone suffices for small moves and fails for large ones.

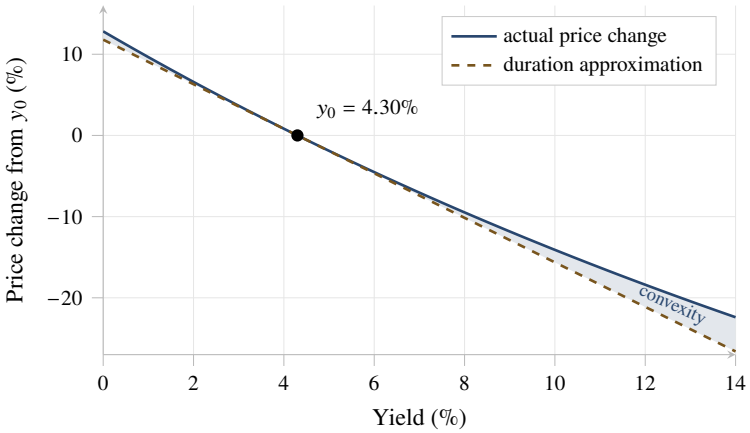


Figure 4.4: Percentage price change against yield for the three-year 5% bond of Example 4.5.

Example 4.5 Duration and convexity of a three-year bond

A three-year bond pays a 5% annual coupon and yields 4.30% with annual compounding. Discounting the cash flows of 5, 5, and 105 gives

$$B = 4.79386 + 4.59623 + 92.54147 = 101.93156$$

The duration from the discrete analog of (4.11), with each cash flow discounted by $(1 + y)^{t_i}$ in

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place of $e^{y t_i}$, is

$$D = \frac{1 \times 4.79386 + 2 \times 4.59623 + 3 \times 92.54147}{101.93156} = \frac{291.61071}{101.93156} = 2.8608 \text{ years}$$

and the modified duration is $D^* = 2.8608/1.043 = 2.7429$. The convexity from (4.13) works out to $C = 10.3499$.

Now suppose the yield rises 50 basis points to 4.80%. The duration-only estimate is

$$\Delta B \approx -101.93156 \times 2.7429 \times 0.005 = -1.39794 \implies B \approx 100.5336$$

Adding the convexity term from (4.14),

$$\frac{1}{2} \times 101.93156 \times 10.3499 \times 0.005^2 = +0.0132 \implies B \approx 100.5468$$

Repricing the bond exactly at 4.80% gives 100.5467. The duration estimate is off by 1.3 cents per 100 of face; adding convexity reduces the error to a hundredth of a cent. For a 50 basis point move on a three-year bond the refinement is marginal—but the convexity term grows with the square of the rate move and with the square of maturity, and on a thirty-year bond in a volatile market it is indispensable.

4.12 THEORIES OF THE TERM STRUCTURE

Why is the zero curve shaped the way it is? Three explanations are traditionally offered.

Expectations theory. Forward rates equal expected future spot rates. An upward-sloping curve then means the market expects rates to rise. This is clean, and Chapter 30 explains precisely why it cannot be exactly right: forward rates are expectations only under a particular probability measure, and the adjustment between that measure and the real world is not zero.

Market segmentation theory. Short, medium, and long maturities are distinct markets with distinct participants (money funds at the short end, pension funds and insurers at the long end), and rates in each are set by local supply and demand. The theory explains persistent curve anomalies that expectations alone cannot, and it received strong support from the 2022 UK gilt episode, in which forced selling by liability-driven investors moved long gilt yields far more than the change in rate expectations.

Liquidity preference theory. Borrowers prefer to lock in long-term funding while lenders prefer to keep money short and available. Clearing that mismatch requires paying lenders a premium to extend, so forward rates exceed expected future spot rates and the curve slopes upward more often than not. This is the explanation most consistent with the historical record: yield curves have been upward-sloping most of the time, which the pure expectations theory would require to mean the market has persistently expected rising rates.

None of the three is complete, and modern practice treats the curve as an observable to be fitted rather than a prediction to be explained. But the intuitions matter when curves behave strangely, and knowing which force is operating is often the difference between recognizing an opportunity and taking a position the market is about to move against.

SUMMARY

The reference rate at the center of the derivatives market has changed, and with it the conventions that everything else is built on. LIBOR was a forward-looking term rate, fixed at the start of each interest period, derived from banks' estimates of their own unsecured funding costs, and therefore carrying an embedded bank credit component. It was manipulated, and the market

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underneath it had largely ceased to exist. It was replaced by overnight rates (SOFR, SONIA, €STR, TONA, SARON), each computed from actual transactions and each close to free of credit risk.

The defining consequence is that a term rate must now be built by compounding daily fixings across an interest period, which means the rate is not known until the period ends. The compounding formula weights each business day's rate by the calendar days it covers and scales the result to the full period, on a day-count base that differs by currency: actual/360 for SOFR and €STR, actual/365 for SONIA and TONA. Because in-arrears determination leaves no time to settle payments, the market developed conventions (payment delay, lookbacks with and without observation shift, lockouts, last-reset structures) that each trade a little economic precision for operational notice, and that each imply slightly different exposures.

Legacy contracts moved to the new rates through the ISDA fallback protocol, which substitutes the compounded RFR plus a spread adjustment fixed permanently in March 2021. Forward-looking term rates exist, built from SOFR futures, but their recommended use is deliberately restricted to loans and their hedges, to avoid recreating LIBOR's vulnerability. Credit-sensitive benchmarks intended to restore banks' funding hedge have largely failed for the same reason, BSBY's discontinuation being the clearest case.

The traditional apparatus survives the transition intact. Rates must always be quoted with a compounding frequency and day count; continuous compounding simplifies the algebra of later chapters. Zero curves are bootstrapped from traded instruments, now principally OIS, and forward rates are implied from them, rising more steeply than the zero curve itself. Duration measures first-order rate sensitivity and convexity the curvature, the latter being both a desirable property of a bond position and, as Chapter 30 develops, a quantity the market prices explicitly.

One simplification made here is corrected in Chapter 7: there is no longer a single curve per currency. Discounting a collateralized trade uses the rate paid on its collateral, while its cash flows may be indexed to something else, so valuation requires a discount curve and one or more forecast curves.

FURTHER READING

Schrimpf and Sushko's "Beyond LIBOR: A Primer on the New Reference Rates" in the March 2019 *BIS Quarterly Review* is a short account of why the benchmarks changed and what replaced them. The Financial Stability Board's progress reports on *Reforming Major Interest Rate Benchmarks* (2014–2023) record the transition as it happened rather than in retrospect. For the mechanics, the Alternative Reference Rates Committee's fallback-language recommendations and its guidance on the scope of use for term SOFR are the operative documents, and Henrard's 2019 note treats the fallback spread methodology in the detail a practitioner needs.

PRACTICE QUESTIONS

4.1 Explain the two distinct defects in LIBOR's construction that led to its replacement, and state which of the two would have justified replacement even if no manipulation had ever occurred.

4.2 An investment pays 12% per annum with quarterly compounding. Give the equivalent continuously compounded and annually compounded rates, and give the semiannual equivalent of a rate quoted as 5% continuously compounded.

4.3 Explain why SOFR can be described as risk-free even though it rose from 2.43% to 5.25% in a single day in September 2019, and state what kind of risk that episode illustrates.

4.4 An interest period runs Monday to Monday, seven calendar days. The overnight rates on Monday through Friday are 3.95%, 3.96%, 3.94%, 3.97%, and 3.96%. Using (4.1),

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calculate the compounded rate on an actual/360 basis and compare it with the day-weighted simple average.

4.5 Explain the difference between a lookback with an observation shift and one without, and why the market generally prefers the shifted convention.

4.6 Explain why it mattered that the ISDA fallback spread adjustments were fixed permanently on a single date rather than recalculated periodically.

4.7 Annual-paying OIS rates are 3.50%, 3.70%, and 3.85% for one, two, and three years. Bootstrap the zero rates and discount factors, calculate the one-year forward rates beginning in one and two years, and explain why an upward-sloping zero curve implies a forward curve that rises more steeply.

4.8 A five-year bond pays a 6% annual coupon and yields 5%. Calculate its price, duration, and modified duration, estimate the price change for a 25 basis point increase in yield, and compare with the exact repricing.

4.9 A forward rate agreement fixes a rate on a notional for a future period. Explain how the forward rate is extracted from two zero rates, why the contract settles at the start of the period rather than at its end, and what that settlement convention requires of the discounting.

4.10 Explain why a single discount curve was adequate before 2008 and is not now. State which curve discounts a collateralized swap and which forecasts its floating payments, and explain why the two being different is a statement about collateral rather than about the credit of the counterparties.

4.11 State the theories of the term structure the chapter sets out, and give what each predicts about the shape of the curve when short rates are expected to stay flat. Explain which prediction the historical record supports, and what that implies for reading a forward rate as a forecast.

FURTHER QUESTIONS

4.12 A bank makes floating-rate loans indexed to term SOFR and funds itself with unsecured deposits. Describe the exposure this creates, explain why LIBOR did not create it, and discuss why credit-sensitive benchmarks intended to solve the problem have largely failed.

4.13 Using the SOFR Index relationship (4.2), show that the compounded rate between any two dates can be obtained from two index values. Explain why this eliminates a class of settlement disputes, and identify what could still be disputed.

4.14 Term SOFR is derived from SOFR futures rather than from submissions. Explain why this makes it more robust than LIBOR, and why regulators nonetheless restricted its recommended scope of use. What would have to be true of the futures market for the restriction to be unnecessary?

4.15 A thirty-year swap written in 2015 referenced three-month USD LIBOR. Describe precisely what it references today, and explain how the value of the contract was preserved through the transition. Under what circumstances would a party have been economically disadvantaged by the fallback?

4.16 Consider a floating-rate note paying compounded SOFR in arrears with a five-day payment delay, and an otherwise identical note using a five-day lookback without observation shift. Explain qualitatively how their values differ, which adjustment from Chapter 30 applies to the first, and why the second cannot be valued exactly from the discount curve.

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